3.2

a) Báze $U = \{u_1, u_2, u_3\} = \{1, x, x^2\}$ na orton. bázi $V = \{v_1, v_2, v_3\}$

Gramovou-Schmidtovou ortogonalizací:

$\tilde{v}_1 = u_1$ $\qquad \|v_1\| = \sqrt{4}$ $\qquad \Rightarrow \boxed{v_1 = \frac{1}{2}} = \frac{\tilde{v}_1}{\|\tilde{v}_1\|}$

$\tilde{v}_2 = a \cdot v_1 + u_2 \qquad / \cdot v_1$

$\underbrace{v_2 \cdot v_1}_{0} = a\, v_1 \cdot v_1 + u_2 \cdot v_1$

$$a = \frac{-u_2 \cdot v_1}{v_1 \cdot v_1} = \frac{-\int_1^5 x \cdot \frac{1}{2}\,dx}{\int_1^5 \frac{1}{2} \cdot \frac{1}{2}\,dx} = \frac{-\left[\frac{x^2}{4}\right]_1^5}{1} = \frac{-\frac{25}{4} + \frac{1}{4}}{1} = -6$$

$\Rightarrow \tilde{v}_2 = -6 \cdot \frac{1}{2} + x = x - 3$ $\qquad \|\tilde{v}_2\| = \sqrt{\int_1^5 (x-3)^2\,dx} = \sqrt{\frac{16}{3}} = \frac{4}{\sqrt{3}}$

$\Rightarrow \boxed{v_2 = \frac{\sqrt{3}}{4} x - \frac{3\sqrt{3}}{4}} = \frac{\tilde{v}_2}{\|\tilde{v}_2\|}$

$\tilde{v}_3 = a \cdot v_1 + b \cdot v_2 + u_3$

i) $/ \cdot v_1$: $\underbrace{\tilde{v}_3 \cdot v_1}_{0} = a \cdot v_1 \cdot v_1 + \underbrace{b \cdot v_2 \cdot v_1}_{0} + u_3 \cdot v_1$

$$a = \frac{-u_3 \cdot v_1}{v_1 \cdot v_1} = \frac{-\int_1^5 x^2 \cdot \frac{1}{2}}{1} = \frac{-\left[\frac{x^3}{6}\right]_1^5}{2} = -\frac{124}{6}$$

ii) $/ \cdot v_2$: $\underbrace{\tilde{v}_3 \cdot v_2}_{0} = \underbrace{a \cdot v_1 \cdot v_2}_{0} + b \cdot v_2 \cdot v_2 + u_3 \cdot v_2$

$$b = \frac{-u_3 \cdot v_2}{v_2 \cdot v_2} = \frac{-\int_1^5 x^2 \cdot \left(\frac{\sqrt{3}}{4} x - \frac{3\sqrt{3}}{4}\right)}{1} = -8\sqrt{3}$$

$\Rightarrow \tilde{v}_3 = \frac{-124}{6} \cdot \frac{1}{2} - 8\sqrt{3} \cdot \left(\frac{\sqrt{3}}{4} x - \frac{3\sqrt{3}}{4}\right) + x^2 = -\frac{124}{12} - \frac{24}{4} x + \frac{24 \cdot 3}{4} + x^2 =$

$= +\frac{23}{3} - \frac{24}{4} x + x^2 = x^2 - 6x + \frac{23}{3}$

$\|\tilde{v}_3\| = \int_1^5 x^2 - 6x + \frac{23}{3}\,dx = \frac{256}{45}$

$\Rightarrow \boxed{v_3 = \frac{45}{256} x^2 - \frac{135}{128} x + \frac{345}{256}} = \frac{\tilde{v}_3}{\|\tilde{v}_3\|}$

$V = \left\{ \frac{1}{2}, \frac{\sqrt{3}}{4} x - \frac{3\sqrt{3}}{4}, \frac{45}{256} x^2 - \frac{135}{128} x + \frac{345}{256} \right\}$

b) Hledáme takové $\tilde{x}$, že $\tilde{x} = P_{\mathrm{span}\{1,x,x^2\}}(\sin x)$, a to metodou nejmenších čtverců a pomocí orton. báze V, $A = (v_1|v_2|v_3)$

$$A^T A [\tilde{x}]_V = A^T \sin x$$

$G_V = I_3$

$$[\tilde{x}]_V = \begin{pmatrix} \langle v_1, \sin x\rangle \\ \langle v_2, \sin x\rangle \\ \langle v_3, \sin x\rangle \end{pmatrix}$$

$$\int_1^5 \frac{1}{2}\sin x \, dx \doteq \boxed{0{,}128 = \langle v_1, \sin x\rangle}$$

$$\int_1^5 \left(\frac{\sqrt{3}}{4}x - \frac{3\sqrt{3}}{4}\right)\sin x \, dx \doteq \boxed{-1{,}493 \doteq \langle v_2, \sin x\rangle}$$

$$\int_1^5 \left(\frac{45}{256}x^2 - \frac{135}{128}x + \frac{345}{256}\right)\sin x \, dx \doteq \boxed{-0{,}053 \doteq \langle v_3, \sin x\rangle}$$

Přejdeme od báze V k bázi U, tedy najdeme ax^2+bx+c

$$\tilde{x} = 0{,}128 \cdot \tfrac{1}{2} - 1{,}493 \cdot \left(\frac{\sqrt{3}}{4}x - \frac{3\sqrt{3}}{4}\right) - 0{,}053 \left(\frac{45}{256}x^2 - \frac{135}{128}x + \frac{345}{256}\right)$$

$$\tilde{x} = -0{,}009x^2 - 0{,}591x + 1{,}934 = ax^2+bx+c$$

c) $$\int_1^5 \left(\sin x - (ax^2+bx+c)\right)^2 dx \doteq 0{,}112$$

$$\sqrt{0{,}112} \doteq 0{,}335 = \| \sin x - (ax^2+bx+c) \|$$